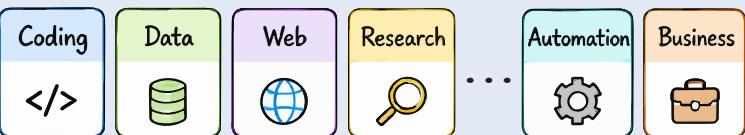


A. Skill-Aware Task Template Building

1 Online Skill Registry

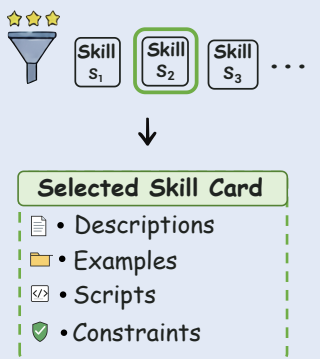


2 Skill Taxonomy

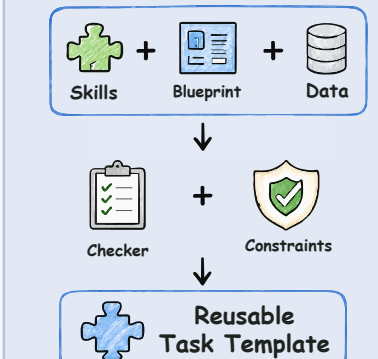


Selected Sub-category

3a High-star Skill Selection

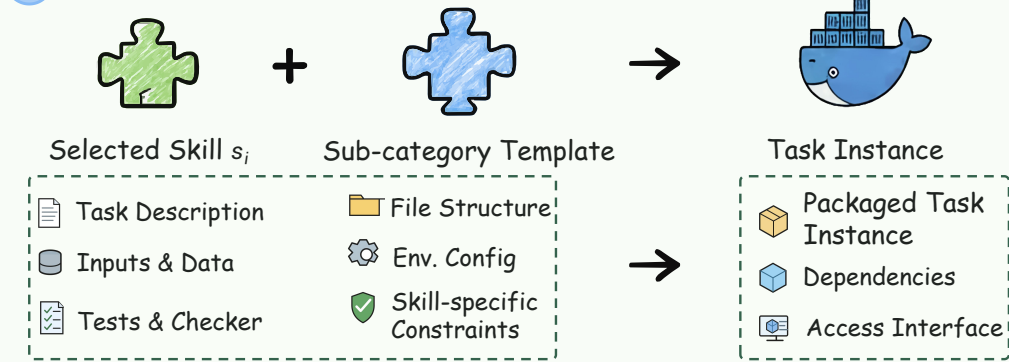


3b Task Template Construction

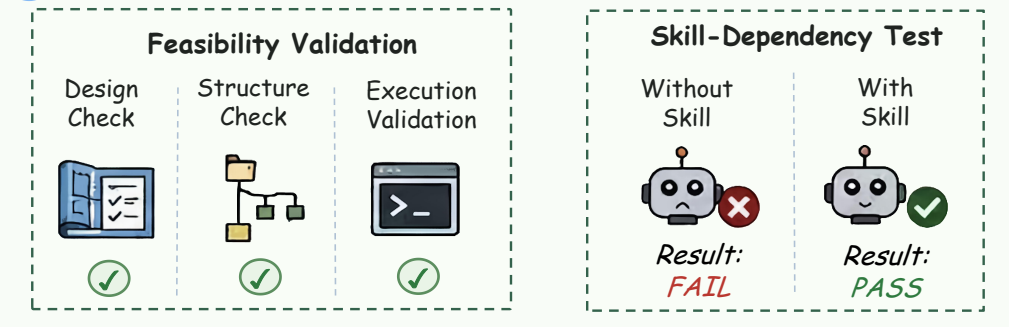


B. Validated Environment Construction

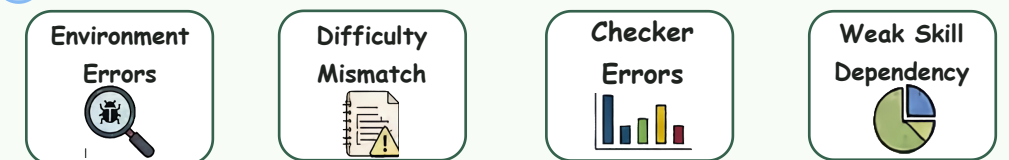
1 Task Construction



2 Failure Analysis & Refinement



3 Validation Suite

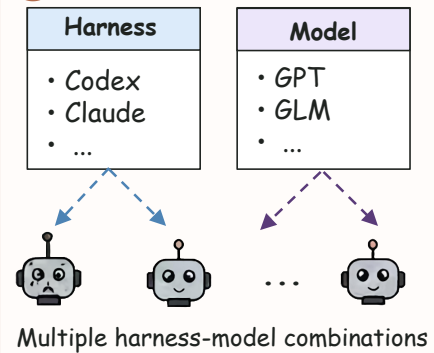


C. Trajectory Sampling

1 Accepted Environment



2 Diverse Trajectory Sampling



3 Sampled Trajectories

